

EX No:1B	EDA – Data Science Frameworks (CRISPDM, SEMMA, KDD)

Aim:

To study Exploratory Data Analysis and apply the CRISP-DM, SEMMA, and KDD data science frameworks to a Life Expectancy dataset by performing data understanding, preprocessing, transformation, modeling, evaluation, and interpretation.

Algorithm:

1. Import pandas, numpy, and the required machine learning libraries.
2. Upload and load the Life Expectancy Data.csv dataset and inspect the first few records and dataset shape.
3. Remove rows with missing values in Life expectancy, drop the Country column, encode categorical variables, and handle missing feature values using mean imputation.
4. Standardize the feature values using StandardScaler and split the dataset into training and testing sets.
5. Train a Random Forest Regressor model to predict Life expectancy and evaluate it using MAE, MSE, RMSE, and R² Score.
6. Apply the CRISP-DM, SEMMA, and KDD frameworks to perform data preparation, modeling, assessment, and interpretation of the Life Expectancy dataset.

Program:

```
import pandas as pd
import numpy as np
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Upload dataset
uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
df.columns = df.columns.str.strip()

print("Dataset Loaded Successfully!")
print(df.head())
print(df.shape)

# =====
# CRISP-DM
# =====
crisp_df = df.copy()

# Business Understanding
# Objective: Predict Life Expectancy
```

```

# Data Understanding
print("\n===== CRISP-DM DATA INFORMATION =====")
print("Dataset Shape:", crisp_df.shape)
print("\nMissing Values:")
print(crisp_df.isnull().sum())

# Data Preparation
crisp_df = crisp_df.dropna(subset=["Life expectancy"])

if "Country" in crisp_df.columns:
    crisp_df.drop("Country", axis=1, inplace=True)

le = LabelEncoder()
for col in crisp_df.select_dtypes(include=["object"]).columns:
    crisp_df[col] = le.fit_transform(crisp_df[col].astype(str))

X = crisp_df.drop("Life expectancy", axis=1)
y = crisp_df["Life expectancy"]

imputer = SimpleImputer(strategy="mean")
X = imputer.fit_transform(X)

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

# Modeling
model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

# Evaluation
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== CRISP-DM RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)

# =====
# SEMMA
# =====
semma_df = df.copy()

# Sample
semma_df = semma_df.dropna(subset=["Life expectancy"])

# Explore
print("\n===== SEMMA DATA INFORMATION =====")
semma_df.info()

# Modify
if "Country" in semma_df.columns:
    semma_df.drop("Country", axis=1, inplace=True)

le = LabelEncoder()
for col in semma_df.select_dtypes(include=["object"]).columns:
    semma_df[col] = le.fit_transform(semma_df[col].astype(str))

X = semma_df.drop("Life expectancy", axis=1)

```

```

y = semma_df["Life expectancy"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

imputer = SimpleImputer(strategy="mean")
X_train = imputer.fit_transform(X_train)
X_test = imputer.transform(X_test)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Model
model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

# Assess
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== SEMMA RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)

# =====
# KDD
# =====
kdd_df = df.copy()

# Step 1: Selection
kdd_df = kdd_df.dropna(subset=["Life expectancy"])

if "Country" in kdd_df.columns:
    kdd_df.drop("Country", axis=1, inplace=True)

# Step 2: Preprocessing
le = LabelEncoder()
for col in kdd_df.select_dtypes(include=["object"]).columns:
    kdd_df[col] = le.fit_transform(kdd_df[col].astype(str))

X = kdd_df.drop("Life expectancy", axis=1)
y = kdd_df["Life expectancy"]

imputer = SimpleImputer(strategy="mean")
X = imputer.fit_transform(X)

# Step 3: Transformation
scaler = StandardScaler()
X = scaler.fit_transform(X)

# Step 4: Data Mining
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

# Step 5: Interpretation
y_pred = model.predict(X_test)

```

```

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== KDD RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)

```

Output:

Dataset Loaded Successfully!

	Country	Year	Status	Life expectancy	Adult Mortality	\
0	Afghanistan	2015	Developing	65.0	263.0	
1	Afghanistan	2014	Developing	59.9	271.0	
2	Afghanistan	2013	Developing	59.9	268.0	
3	Afghanistan	2012	Developing	59.5	272.0	
4	Afghanistan	2011	Developing	59.2	275.0	

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	\
0	62	0.01	71.279624	65.0	1154	...	
1	64	0.01	73.523582	62.0	492	...	
2	66	0.01	73.219243	64.0	430	...	
3	69	0.01	78.184215	67.0	2787	...	
4	71	0.01	7.097109	68.0	3013	...	

	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	\
0	6.0	8.16	65.0	0.1	584.259210	33736494.0	
1	58.0	8.18	62.0	0.1	612.696514	327582.0	
2	62.0	8.13	64.0	0.1	631.744976	31731688.0	
3	67.0	8.52	67.0	0.1	669.959000	3696958.0	
4	68.0	7.87	68.0	0.1	63.537231	2978599.0	

	thinness	1-19 years	thinness	5-9 years	Income composition of resources	\
0		17.2		17.3		0.479
1		17.5		17.5		0.476
2		17.7		17.7		0.470
3		17.9		18.0		0.463
4		18.2		18.2		0.454

	Schooling
0	10.1
1	10.0
2	9.9
3	9.8
4	9.5

```

[5 rows x 22 columns]
(2938, 22)

```

===== CRISP-DM DATA INFORMATION =====

Dataset Shape: (2938, 22)

Missing Values:

Country	0
Year	0
Status	0
Life expectancy	10
Adult Mortality	10
infant deaths	0
Alcohol	194
percentage expenditure	0
Hepatitis B	553
Measles	0
BMI	34
under-five deaths	0
Polio	19
Total expenditure	226
Diphtheria	19
HIV/AIDS	0
GDP	448
Population	652
thinness 1-19 years	34
thinness 5-9 years	34
Income composition of resources	167
Schooling	163

dtype: int64

===== CRISP-DM RESULTS =====

MAE : 1.0588054607508561
MSE : 2.7685972969283315
RMSE : 1.6639102430504873
R2 Score : 0.9679928736778992

===== SEMMA DATA INFORMATION =====

<class 'pandas.core.frame.DataFrame'>
Index: 2928 entries, 0 to 2937

```

===== SEMMA DATA INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
Index: 2928 entries, 0 to 2937
Data columns (total 22 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Country                                   2928 non-null   object
1   Year                                     2928 non-null   int64
2   Status                                   2928 non-null   object
3   Life expectancy                         2928 non-null   float64
4   Adult Mortality                        2928 non-null   float64
5   infant deaths                          2928 non-null   int64
6   Alcohol                                 2735 non-null   float64
7   percentage expenditure                 2928 non-null   float64
8   Hepatitis B                           2375 non-null   float64
9   Measles                                2928 non-null   int64
10  BMI                                    2896 non-null   float64
11  under-five deaths                     2928 non-null   int64
12  Polio                                 2909 non-null   float64
13  Total expenditure                     2702 non-null   float64
14  Diphtheria                            2909 non-null   float64
15  HIV/AIDS                              2928 non-null   float64
16  GDP                                    2485 non-null   float64
17  Population                            2284 non-null   float64
18  thinness 1-19 years                   2896 non-null   float64
19  thinness 5-9 years                    2896 non-null   float64
20  Income composition of resources       2768 non-null   float64
21  Schooling                             2768 non-null   float64
dtypes: float64(16), int64(4), object(2)
memory usage: 526.1+ KB
None

```

```

===== SEMMA RESULTS =====
MAE : 1.0500972696245772
MSE : 2.768727247440278
RMSE : 1.6639492923284285
R2 Score : 0.9679913713494616

```

```

===== KDD RESULTS =====
MAE : 1.0588054607508561
MSE : 2.7685972969283315
RMSE : 1.6639102430504873
R2 Score : 0.9679928736778992

```

Result:

